Shahkar Ahmad Shah



BS Computer Science (Artificial Intelligence), University of Peshawar · Applying for a research Master's in China
Large Language Models; LLM fine-tuning and adaptation; Retrieval-Augmented Generation (RAG); LLM reasoning and evaluation; reliable multi-agent LLM systems; hallucination detection and fact verification
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Portfolio | GitHub | LinkedIn

EDUCATION

BS Computer Science (AI Specialisation) — University of Peshawar, Pakistan Oct 2022 – 2026
CGPA 3.13 / 4.00 · All graduation requirements completed; degree conferral pending

Advanced AI Bootcamp (7 weeks) — GIK Institute Aug 2026
Transformers & LLM fine-tuning, post-training & alignment (PEFT, RLHF), agentic RAG, LLMOps

FSc — Government College Peshawar · 998 / 1100 (90.7%), BISE Peshawar 2022

English: IELTS Academic scheduled for December 2026

RESEARCH PROJECTS

MootCourtSimulator — Multi-Agent LLM System with Verified Legal Retrieval 2026
*Bootcamp capstone, GIK Institute · **Honorable Mention, Top 5 of 35** · LangGraph, OpenAI API, Whisper, FastAPI, MLflow*

- Voice-first Built a virtual courtroom using LangGraph, where multiple AI agents (Judge, Opposing Counsel, and Witness) work together to simulate legal proceedings.
- Developed a hybrid retrieval system combining BM25 keyword search and semantic search, using Reciprocal Rank Fusion (RRF) to find relevant legal documents.
- Integrated LLM-based reranking and legal citation verification to help ensure that responses are supported by the Qanun-e-Shahadat Order 1984.
- Evaluated the system on 32 test scenarios, achieving MRR 1.00 and Objection F1 1.00. Received an Honorable Mention (Top 5/35) at the GIKI AI Bootcamp.

Text-to-SQL with Reinforcement Learning (GRPO) — Execution-Verified Rewards 2026
Independent project · GitHub · PyTorch, Hugging Face TRL, LoRA, SQLite, Gradio

- Fine-tuned Qwen2.5-Coder-3B using GRPO (Group Relative Policy Optimization) to convert natural-language questions into SQL queries.
- Used execution-based rewards to evaluate generated SQL by running queries against SQLite databases and checking whether their results matched the reference answers.
- Applied LoRA (Low-Rank Adaptation) to fine-tune the model efficiently on a single 16GB GPU.
- Improved execution accuracy from 64.5% to 67.6% on Spider and 32.6% to 38.2% on BIRD Mini-Dev.

AI Content Factory — Evaluation-Gated Multi-Agent Content Generation 2025 – 2026
Final Year Project (team of 2), supervisor Dr. Syed Muhammad Asim Jalal · GitHub · LangGraph, LangChain, ChromaDB, DeepEval

- Developed a 10-agent LangGraph system that uses AI agents to research a topic and generate blog posts, social media content, podcasts, and videos.
- Integrated Retrieval-Augmented Generation (RAG) to provide agents with relevant source material when generating content.
- Used DeepEval and G-Eval to evaluate content quality, relevance, and factual grounding, alongside an additional citation verification system.
- Implemented human approval of content outlines and separate research material for parallel writing agents to reduce duplicated content

RESEARCH COLLABORATION

Research Collaborator (remote) — ML for Molecular Property Prediction 2025 – Present
With Muhammad Amjad, Master's student, Department of Pharmacy, GNU, South Korea

- Predicted the enhancement/penetration ratio of 70 surfactant compounds at two concentrations (140 samples) from RDKit descriptors: $R^2 = 0.56$, MAE = 0.09 vs 0.17 for a mean-value baseline.

EXPERIENCE

AI Engineer Intern (remote, part-time) — The Nerd Camp, Peshawar Jan – May 2026
Five-month internship building AI and ML-based applications; recognised by the CEO; experience letter available on request.

TECHNICAL SKILLS

LLMs & agents: LangGraph, LangChain, OpenAI API, Ollama, Multi-Agent Systems
LLM Fine-Tuning: LoRA, QLoRA, GRPO, Hugging Face PEFT, TRL
RAG & Information Retrieval : RAG, BM25, Dense Retrieval, RRF, LLM Reranking, FAISS, ChromaDB
Backend & Tools: Python, SQL, FastAPI, Gradio, Git

PERSONAL & REFERENCES

Nationality: Pakistani · **Date of birth:** 12 April 2004 · **Languages:** Urdu, Pashto (native); English (professional)
References: Dr. Syed Muhammad Asim Jalal (FYP supervisor) and Dr. Rahman Ali, Assistant Professors, Dept. of Computer Science, University of Peshawar · Letters: Dr. Jalal, Dr. Ali